



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

Advance **IoT Based** Remote Patient **Health Monitoring System**

Course: **MTE 3104** – Microcontroller and Interfacing Sessional

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WHAT IS BROKEN?

According to the Bangladesh Bureau of Statistics,

Deaths In Rural Bangladesh that Could Be Addressed Without Timely Healthcare Monitoring

60%



Delays in Receiving Critical Health Information

70%



Lack Access To Quality Healthcare Monitoring

40%

WHY IS IT BROKEN?



Limited Facilities

Skilled personnel & Internet hinder Monitoring



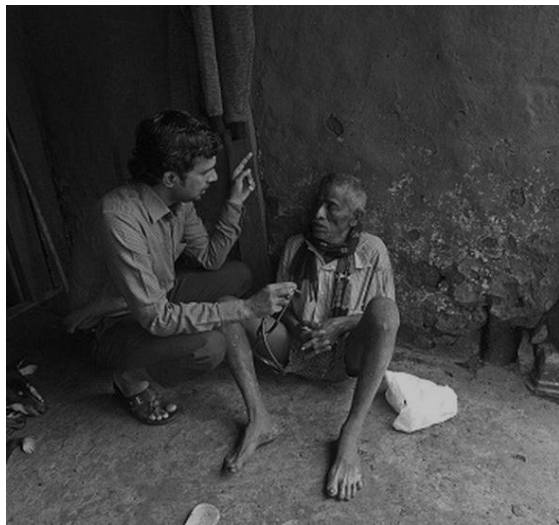
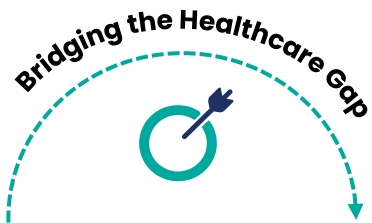
Limited Access

Healthcare Infrastructure & Monitoring Devices



SDG Alignment

SOLUTION



Healthcare Challenges



Wireless Patient Monitoring System



Improved Healthcare Outcomes



PROJECT IDEA

A wireless patient health monitoring system through
mobile app - “সচেতন”



MAX30100 Pulse
Oximeter Sensor



DS18B20
Temperature Sensor



DHT11
Humidity Sensor



Connecting Wires

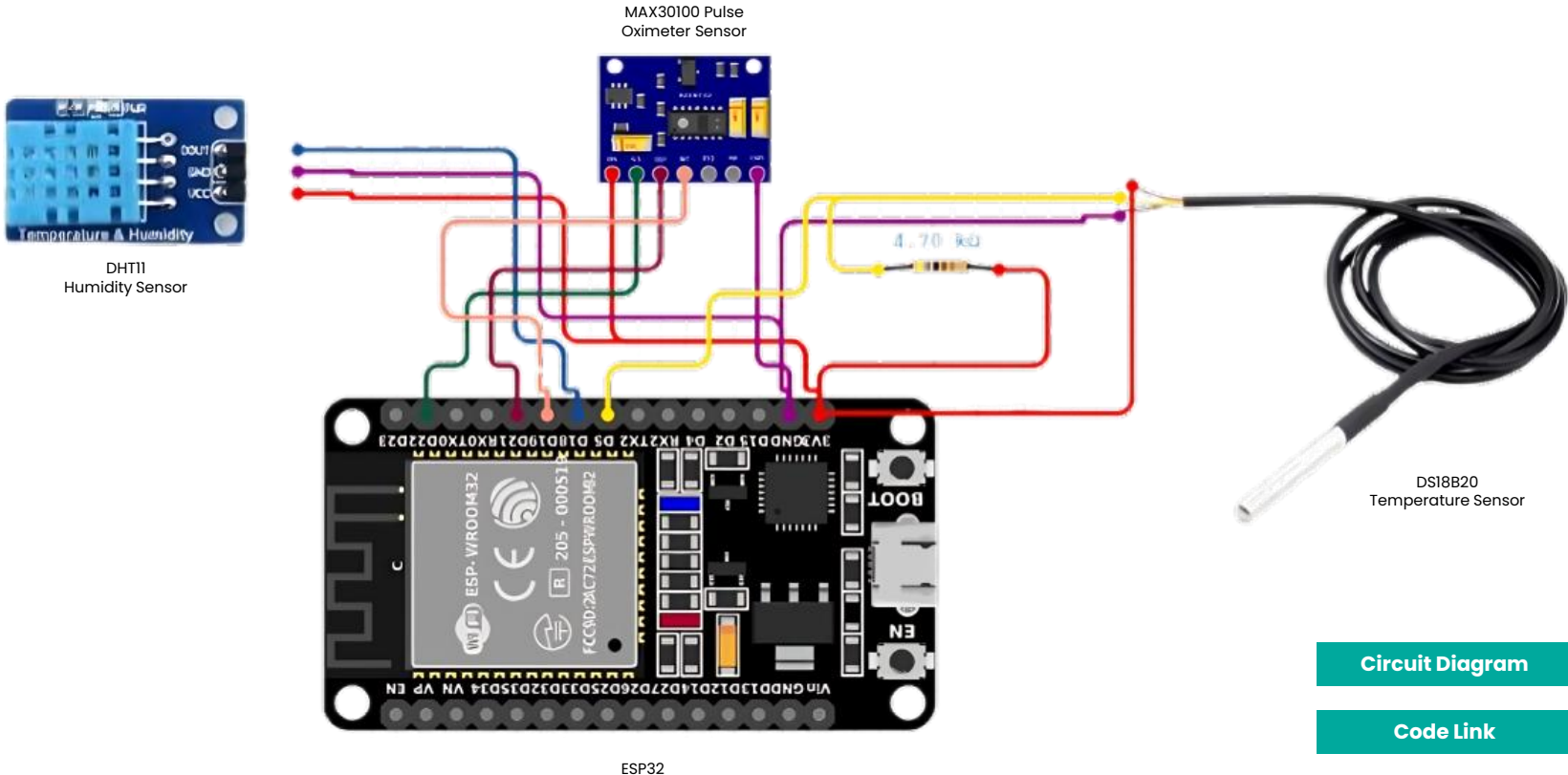


ESP32



সচেতন

CIRCUIT DIAGRAM



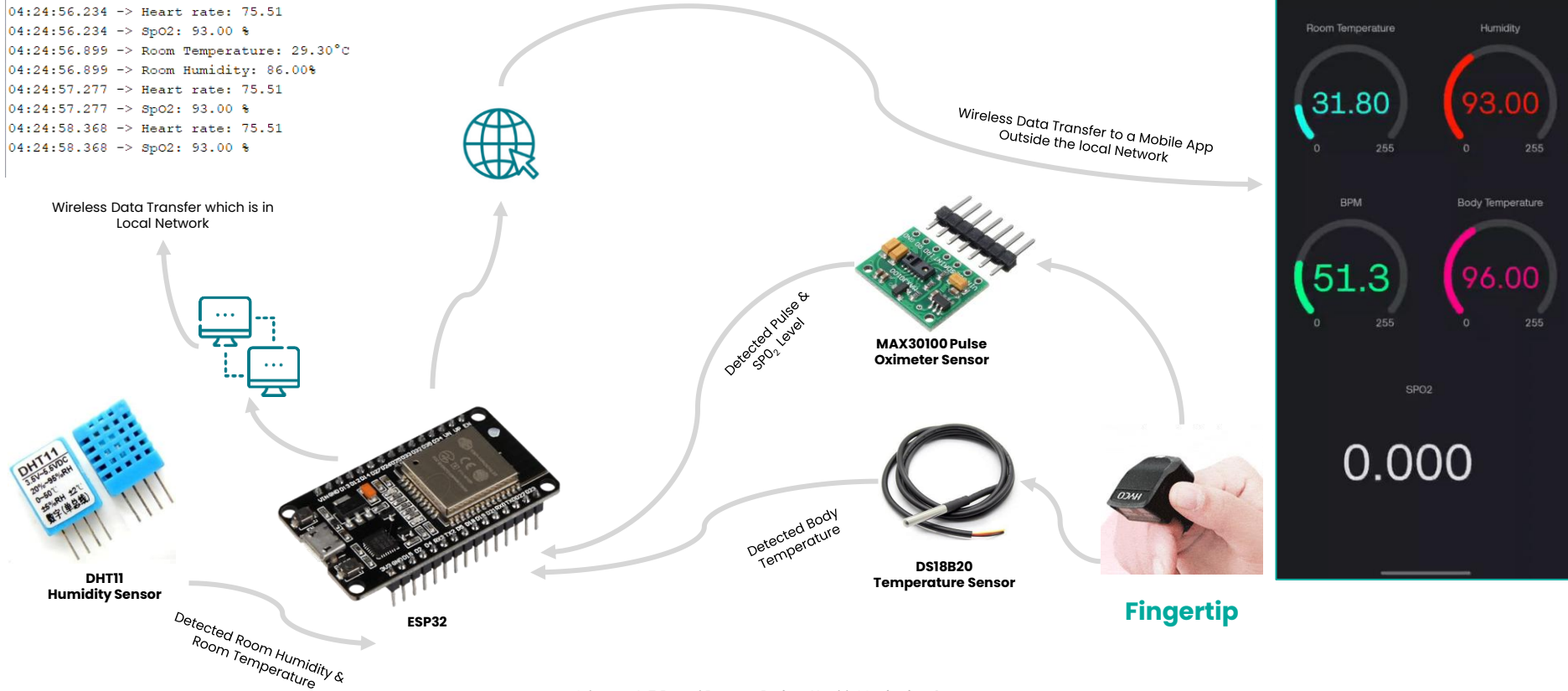
[Circuit Diagram](#)

[Code Link](#)

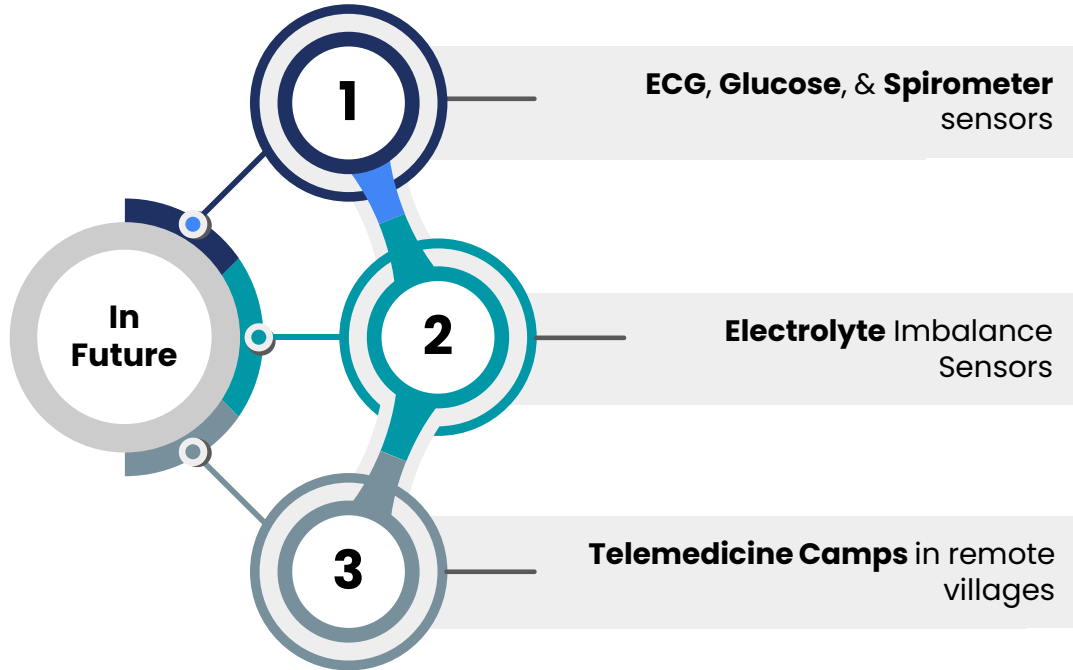
ESP32

How “সচেতন” Works

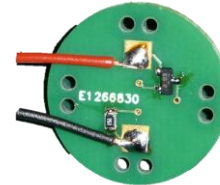
```
04:24:54.099 -> Heart rate: 75.51
04:24:54.099 -> SpO2: 93.00 %
04:24:54.903 -> Room Temperature: 29.30°C
04:24:54.903 -> Room Humidity: 86.00%
04:24:55.141 -> Heart rate: 75.51
04:24:55.141 -> SpO2: 93.00 %
04:24:56.234 -> Heart rate: 75.51
04:24:56.234 -> SpO2: 93.00 %
04:24:56.899 -> Room Temperature: 29.30°C
04:24:56.899 -> Room Humidity: 86.00%
04:24:57.277 -> Heart rate: 75.51
04:24:57.277 -> SpO2: 93.00 %
04:24:58.368 -> Heart rate: 75.51
04:24:58.368 -> SpO2: 93.00 %
```



FUTURE OPPORTUNITIES



ECG Sensor



Electrolyte Imbalance Sensor



Spirometer Sensor



Telemedicine Camps in village schools



Glucose Sensor



**Let's Connect the Healthcare
Solution with just a
fingertip.**

**THANK
YOU**

APPENDIX

PROOF OF CONCEPT

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Chapter PDF Available

IoT Based Health Monitoring System

November 2021

DOI:10.3233/AFG210140

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In book: Advances in Parallel Computing Technologies and Applications

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Article PDF Available

A Health Monitoring System Using IoT-Based Android Mobile Application

January 2023 - Computer Systems Science and Engineering 47(2):2293-2311

DOI:10.32604/csse.2023.040312

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Admin (2023) *IoT based patient health monitoring on ESP32 web server, How To Electronics*. Available at: <https://how2electronics.com/iot-based-patient-health-monitoring-esp32-web-server/> (Accessed: 18 October 2024).

Component List

S.N.	Components Name	Quantity	Price
1	ESP32 Board	1	480 Tk
2	MAX30100 Pulse Oximeter Sensor	1	350 Tk
3	DS18B20 Sensor	1	150 Tk
4	DHT11 Sensor	1	200 Tk
5	Resistor 4.7K	1	5 TK
6	Connecting Wires	10	50 Tk
7	Breadboard	1	70 Tk
Total:			1305 Tk